LETTER

doi:10.1038/nature25988

Image reconstruction by domain-transform manifold learning

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Received 12 April 2017; accepted 23 January 2018.

https://doi.org/10.1038/nature25988

Presented on April 10th, 2021

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Introduction

What is the problem?

- Standard image reconstruction involves approximating the inverse function with multiple *ad hoc* stages in a signal processing chain, which depends on the details of each acquisition strategy, and expert parameter tuning.

Why is this important?

- This work presents a unified framework for image reconstruction which recasts image reconstruction as a data-driven supervised learning task that allows a mapping between the sensor and the image domain to emerge from an appropriate corpus of training data.

Introduction

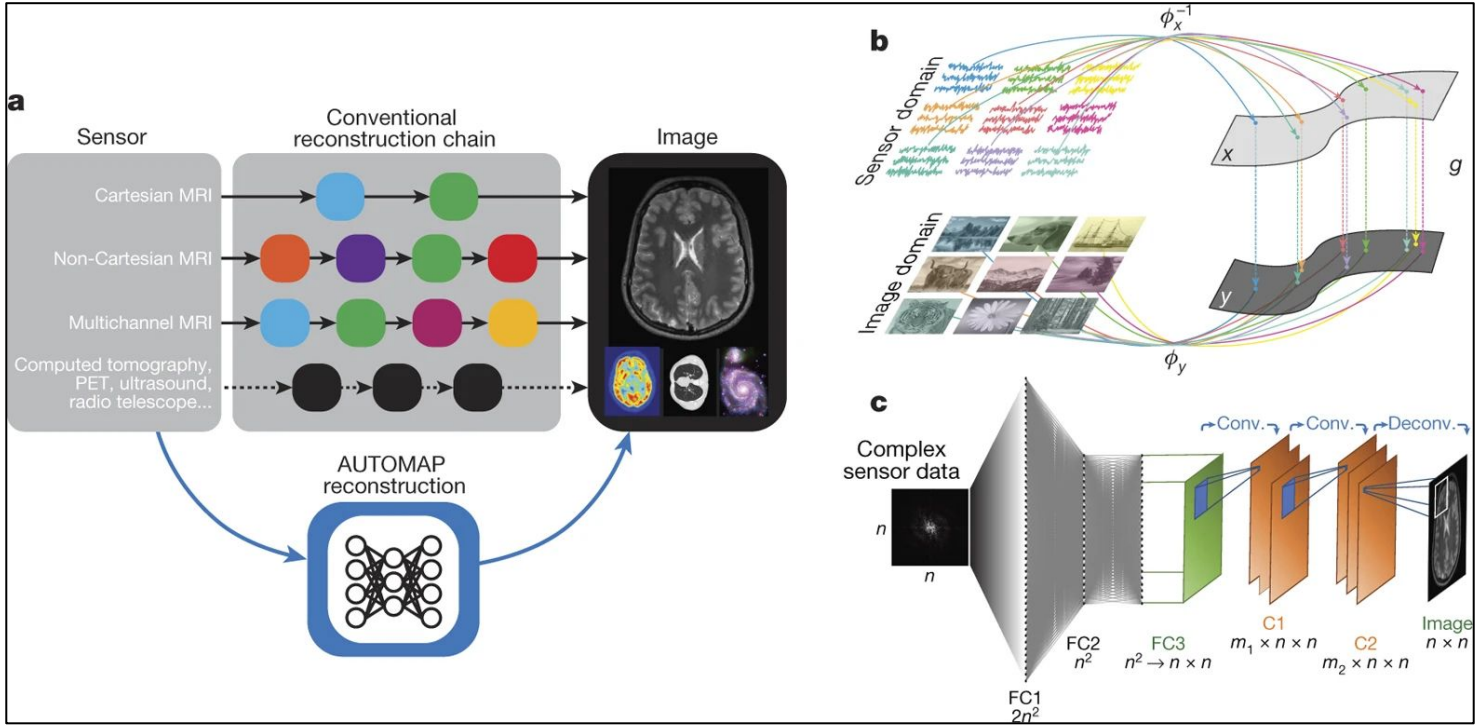
What have the authors done differently to solve this problem?

- Implemented automated transform by manifold approximation (**AUTOMAP**) with a deep neural network and exhibit its flexibility in learning reconstruction transforms for various MRI acquisition strategies, using the same network architecture and hyperparameters.

Results

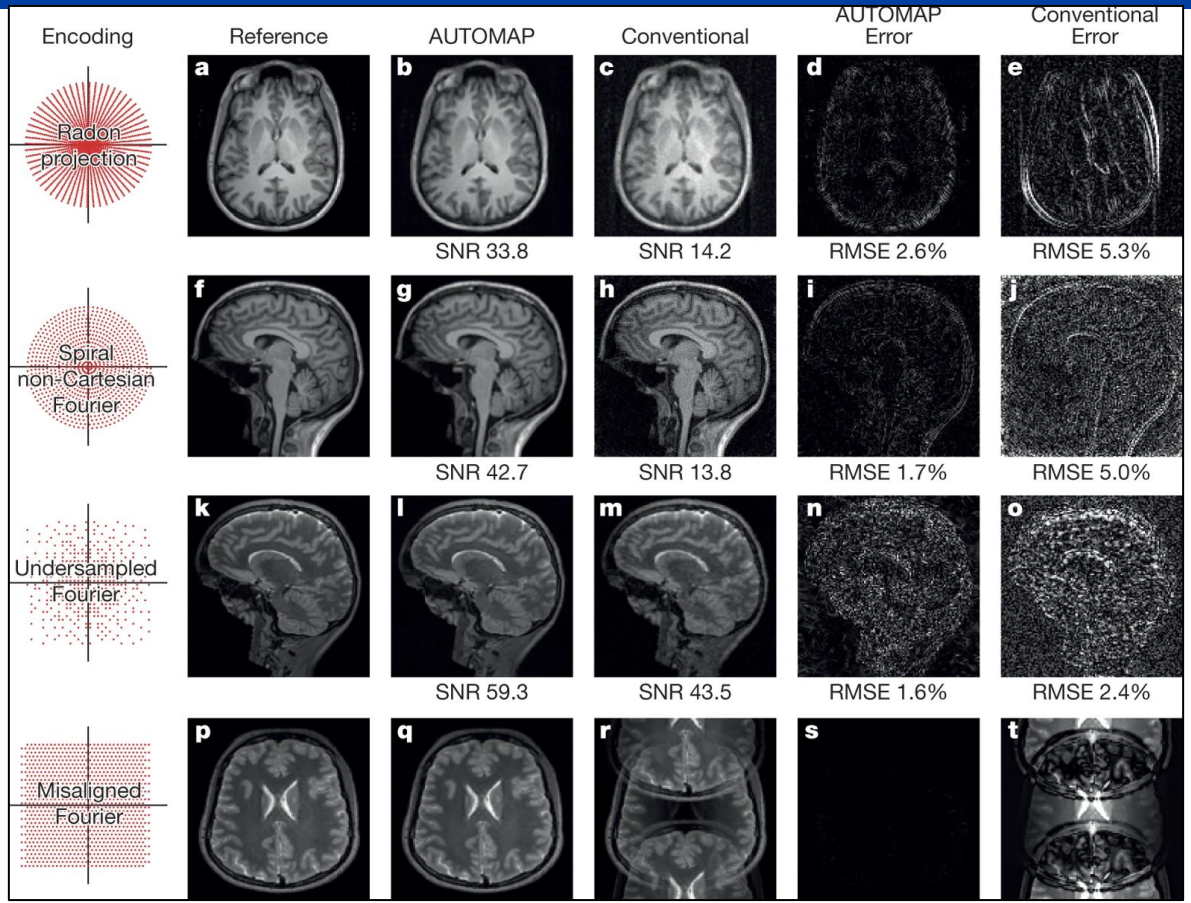
- Manifold learning during training results in a sparse representations of domain transforms along low-dimensional data manifolds, and observe superior immunity to noise and a reduction in reconstruction artefacts.

Schematic Representation



- Learns a reconstruction mapping between the sensor-domain data and image-domain output.
- As this mapping is trained, a low-dimensional joint manifold of the data in both domains is learned.
- Deep neural network feedforward architecture composed of fully connected layers followed by a sparse convolutional autoencoder.

Reconstruction Performance



- Exhibit superior noise immunity compared to other conventional methods.

Sensor-domain encoding

Sensor-domain representations for each image were encoded according to the reconstruction task.

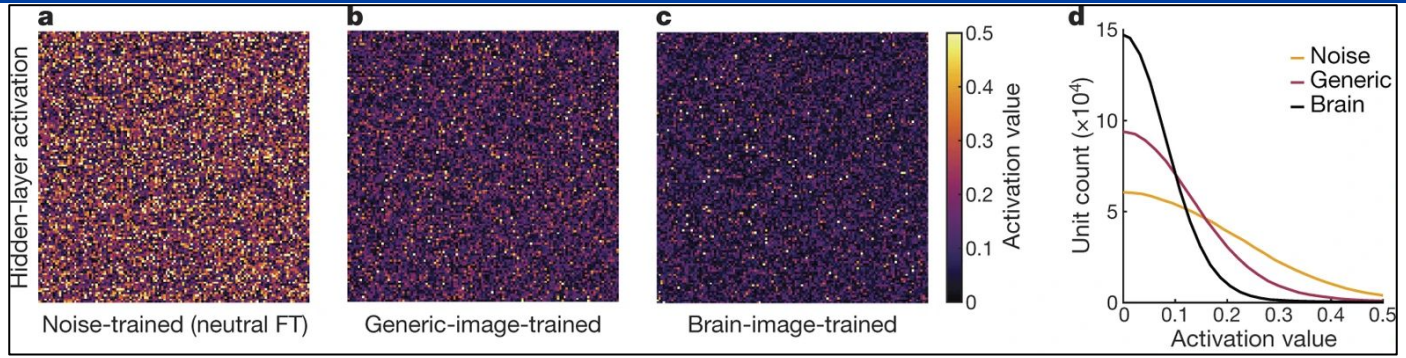
- Radon transform experiment (Fig. 2a–c), we used the discrete Radon transform with 180 projection angles and 185 parallel rays.

- The spiral k-space experiment (Fig. 2d–f) used nonuniform fast Fourier transform (NUFFT) [**3**] to encode a ten-interleave spiral trajectory [**4**] with variable density factor $\alpha = 1$ and undersampling factor $R = 1/1.2$ based on the pre-subsampled 256×256 images (the MATLAB code used for this trajectory encoding is available at http://bigwww.epfl.ch/algorithms/ mri-reconstruction/).

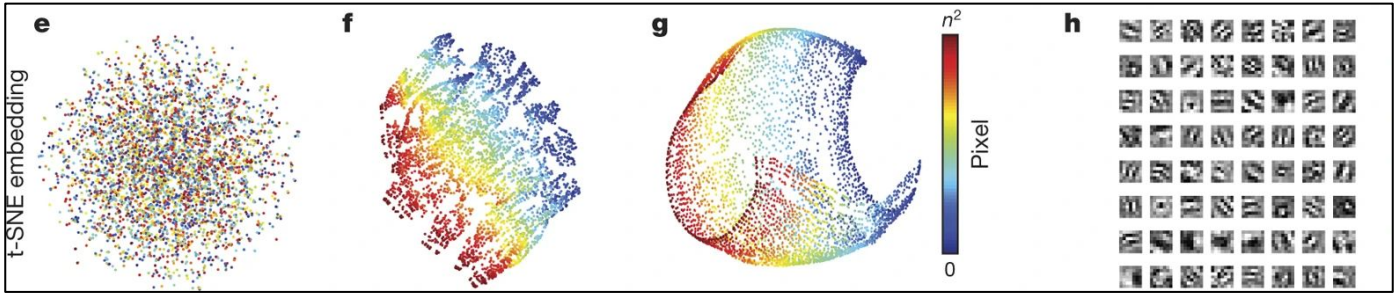
- The undersampled Cartesian k-space experiment (Fig. 2g–i) used a Poisson-disk sampling pattern with 40% undersampling of the Fourier-transformed k-space generated with the Berkeley Advanced Reconstruction Toolbox (BART) [**5**].

- The misaligned Cartesian k-space experiment (Fig. 2j–l) used Fourier-transformed k-space where an empirically observed phase shift from a typical echo planar imaging acquisition was applied to every odd readout line.

Neural Network Analysis

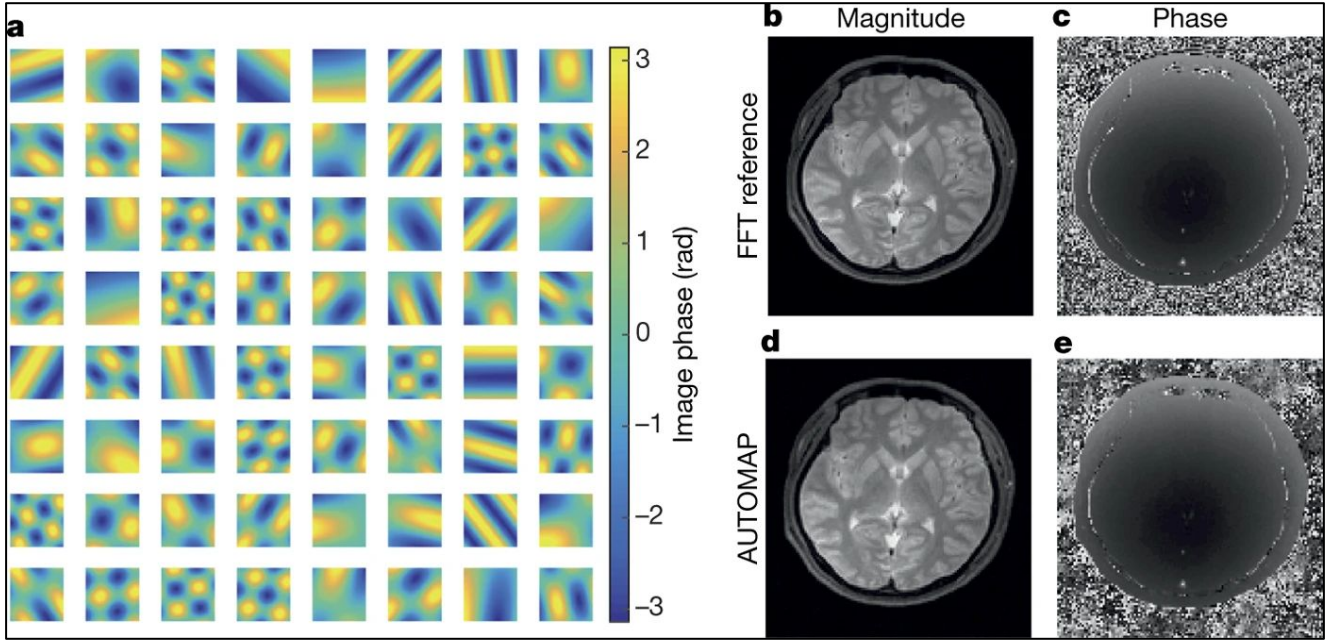


- As the training moves from general to specific (a → c), the hidden-layer activity exhibiting greater sparsity is observed, indicating successful extraction of robust features.
- (**d**) The t-SNE embedding of the noise-trained network weights is highly unorganized and unbiased with respect to pixel location.



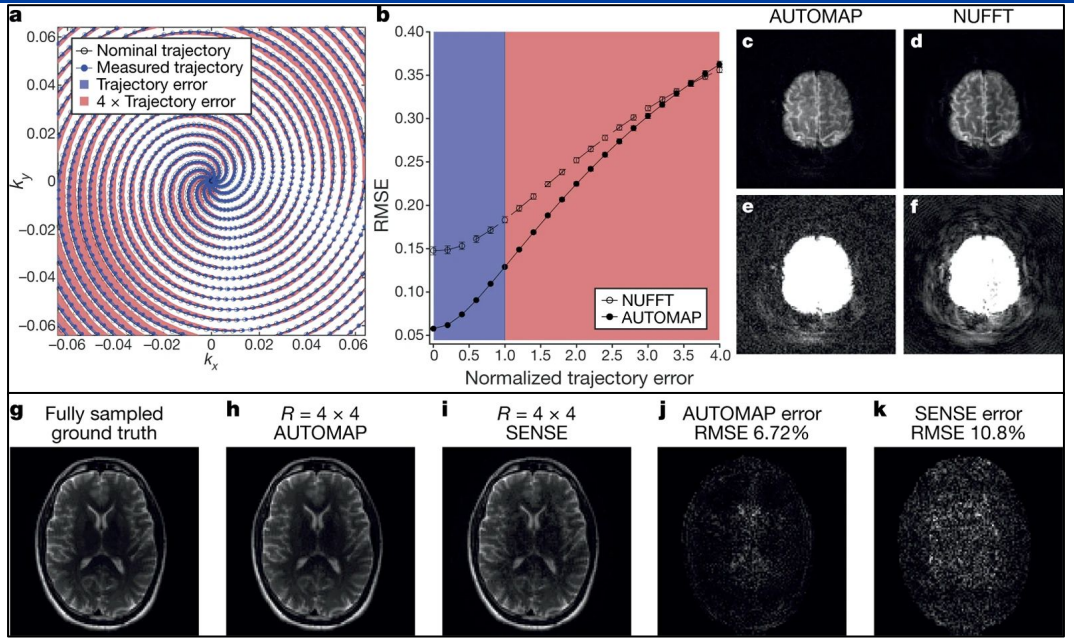
- (**f**, **g**) The generic-image-trained model captures this local spatial correlation, so the weights of neighbouring pixels are more similar than the weights of pixels further apart.

Learning Reconstruction of Phase



- (**a**) A phase-modulated training set was created by generating synthetic phase patterns to modulate the magnitude-only training images collected from the HCP database before being encoded in k-space.
- Using the same k-space data as input, we trained separate AUTOMAP networks to reconstruct magnitude and phase with their respective target training images, and validated this by reconstructing in vivo k-space raw data taken from a human subject on a 3T-MRI scanner.

Performance in Real-World Acquisitions



- (**a**) Quantified the effect of divergence from the nominal sampling trajectory with Monte Carlo simulations of varying amplitudes of trajectory error from a spiral acquisition, and measure the resulting reconstruction RMSE from the ground truth reference.
- (**b**) AUTOMAP's reconstruction error smoothly increased as a function of trajectory error, similar to a conventional NUFFT reconstruction's error curve, demonstrating robustness to trajectory deviation.
- (**c**) Reconstructions of a 10-interleave spiral MR acquisition - AUTOMAP output had a higher image SNR.

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